

Madiha Khan

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EDUCATION

University of Calgary

Calgary, AB

Bachelor of Science in Software Engineering

Sept. 2023 – April 2027

- **Relevant Courses:** Networked Systems, Data Management, Embedded Software & Hardware Systems, Object-Oriented Programming, Data Structures & Algorithms, Full Stack Web Development, Computer Organization, Applied Operating Systems, Software Testing & Architecture, Software Performance Evaluation, Project Management, Machine Learning.
- **Availability:** Available for internships starting May 1, 2026.

EXPERIENCE

Chorate AI

Remote

Software Engineering Internship

Dec. 2025 – Mar. 2026

- Engineered real-time system monitoring dashboards by integrating OS-level metrics (*psutil*) with a web frontend to proactively track resource utilization and server health.
- Assisted in tasks and feature development assigned by engineering supervisors in a dynamic, remote team environment.

Calgary Board of Education

Calgary, AB

Casual Lunch Supervisor

May 2023 – Aug. 2025

- Monitored safety protocols and managed complex group dynamics efficiently in a fast-paced environment.
- Communicated effectively with multi-disciplinary team members to ensure smooth, safe daily facility operations.

PROJECTS

LastSeen - Room-memory Mobile App | *Expo, React Native, FastAPI, Supabase, OpenAI, Gemini* May 2026

- Developed a mobile application using Expo and React Native to capture room frames and record visual events through a custom change-detection pipeline.
- Integrated a FastAPI backend with Supabase and AI synthesis framework (OpenAI/Gemini) to process complex natural-language queries for spatial item localization.
- Won 4th place at the Cursor Calgary Hackathon with a high-fidelity, demo-ready prototype featuring optimized image handling and a polished UX/UI layout.

Ice Cream Website | *React, Flask, REST APIs, bcrypt, Python, MySQL* Jan. 2026 – Apr. 2026

- Built a full-stack e-commerce ordering web application connecting user signup, secure login, persistent shopping carts, and past order histories via custom REST APIs.
- Implemented bulletproof authentication workflows using client-side session persistence, secure protected application routes, and backend bcrypt password hashing.
- Developed robust backend validation logic and reusable modular frontend components to handle rapid product browsing, cart updates, and checkouts smoothly.

Flight Reservation System | *Java, SQL, JDBC, Object-Oriented Programming* Oct. 2025 – Dec. 2025

- Designed and developed a standalone booking software platform with a dedicated focus on secure user data flows and clean database persistence layout.
- Implemented user authentication states, multi-criteria flight search filters, and an administrative booking ledger management subsystem.
- Structured application architecture for maximum scalability and structural maintainability adhering strictly to modular Java development patterns.

Count Down Timer App | *C, Python, Embedded Systems, Driver Architecture* Oct. 2025

- Programmed a hardware microcontroller using C and Python script environments to establish a precise real-time countdown tracking engine.
- Managed foundational physical hardware layer interactions, customized low-level device driver functions, and benchmarked bugs using professional utilities like RealTerm and MPLab.

EXTRACURRICULARS

Knowledge Bridge Engineering Club

Calgary, AB

Executive

May 2026 – Present

- Sustained active educational research bridging the strict gap between technical university course lectures/labs and functional, real-world tech industries.
- Bridges foundational student experience expectations directly with engineering department course learning outcomes.

MISS UCalgary Club

Calgary, AB

VP of External Affairs

May 2024 – Dec. 2025

- Collaborated tightly with corporate industry sponsors and internal cross-functional university teams to secure financial project funding and design large scale professional networking events.
- Demonstrated excellent remote organizational coordination, clear communication, and high adaptability in a modern hybrid operational ecosystem.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL, HTML5, CSS3, MATLAB

Frameworks & Libraries: React.js, Next.js, React Native, Expo, Expo Router, Metro Bundler, Node.js, FastAPI, Flask, Pydantic, TailwindCSS

Developer Tools: Git, GitHub, Docker, Kubernetes, Linux/Unix Shell, npm, Visual Studio, VS Code, Cursor, RealTerm, MPLab, Microsoft Office Suite

Databases & Storage: MySQL, PostgreSQL, Supabase

Concepts & Methodologies: Object-Oriented Design, Data Structures & Algorithms, Embedded Systems, Scripting, Workflow Automation, Debugging, Performance Optimization, UI/UX Implementation, Remote Collaboration